SolarMood

SDG 9

A live pulse of how the world's news talks about solar energy for Holland Solar's policy team

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Minor AI for Good · Hackathon 6 · De Haagse Hogeschool

The Dutch solar sector is under pressure

−56%

new solar capacity in NL in 2025 (1,447 MWp vs ~3.3 GWp in 2024)

1 Jan 2027

net metering (salderingsregeling) ends for ~3.3 million installations

~180 / day

English solar headlines our tool found, from 221 outlets in 54 hours

! **What goes wrong today:** nobody can read 180 international headlines a day. Themes like Chinese overcapacity, panel fires or subsidy cuts reach the Dutch debate before the sector association has a position ready. So it reacts instead of preparing.

Sources: CBS via Solar Magazine (13 Jun 2026); Solar Magazine (17 Dec 2024), Eerste Kamer vote on net metering; own data: solarmood.csv, 12–14 Oct 2026

Our user: Holland Solar's policy team

Who & what they decide

- Dutch trade association for the solar sector since 1983 (Utrecht), speaking for installers, manufacturers and developers
- **Weekly decision: which theme gets a position, member update or press line this week?**
- Needs: a 5-minute weekly scan, the headlines behind every dip, no programming
- Benefits: member companies in a sector of ~36,000 jobs, and indirectly 3.3 million panel owners

Not our users

- Investors deciding on solar stocks
- Consumers deciding whether to buy panels
- Anyone judging one company or person

Missing from the stream

- Dutch-language news (feed is en-US)
- Residents near solar parks and people who never post or get quoted
- Social media: we read headlines, not opinions

Sources: hollandsolar.nl/leden; SolarPower Europe EU Solar Jobs Report 2022 (36,444 NL jobs, 2021); CBS via Solar Magazine (13 Jun 2026)

How SolarMood works: from feed to dashboard

1

Fetch

Google News RSS, 4 search terms (solar energy / panels / power / technology), last 24 h

2

Repeat

auto_collector.py runs collector.py every 30 minutes, no button

AI

Analyse

Pre-trained RoBERTa labels each new headline: positive / neutral / negative + confidence

4

Store

solarmood.csv: headline, publish time, label, confidence, collected_at. Duplicates skipped

5

Show

Streamlit dashboard: totals, sentiment per hour, filter by label and hour to open the headlines

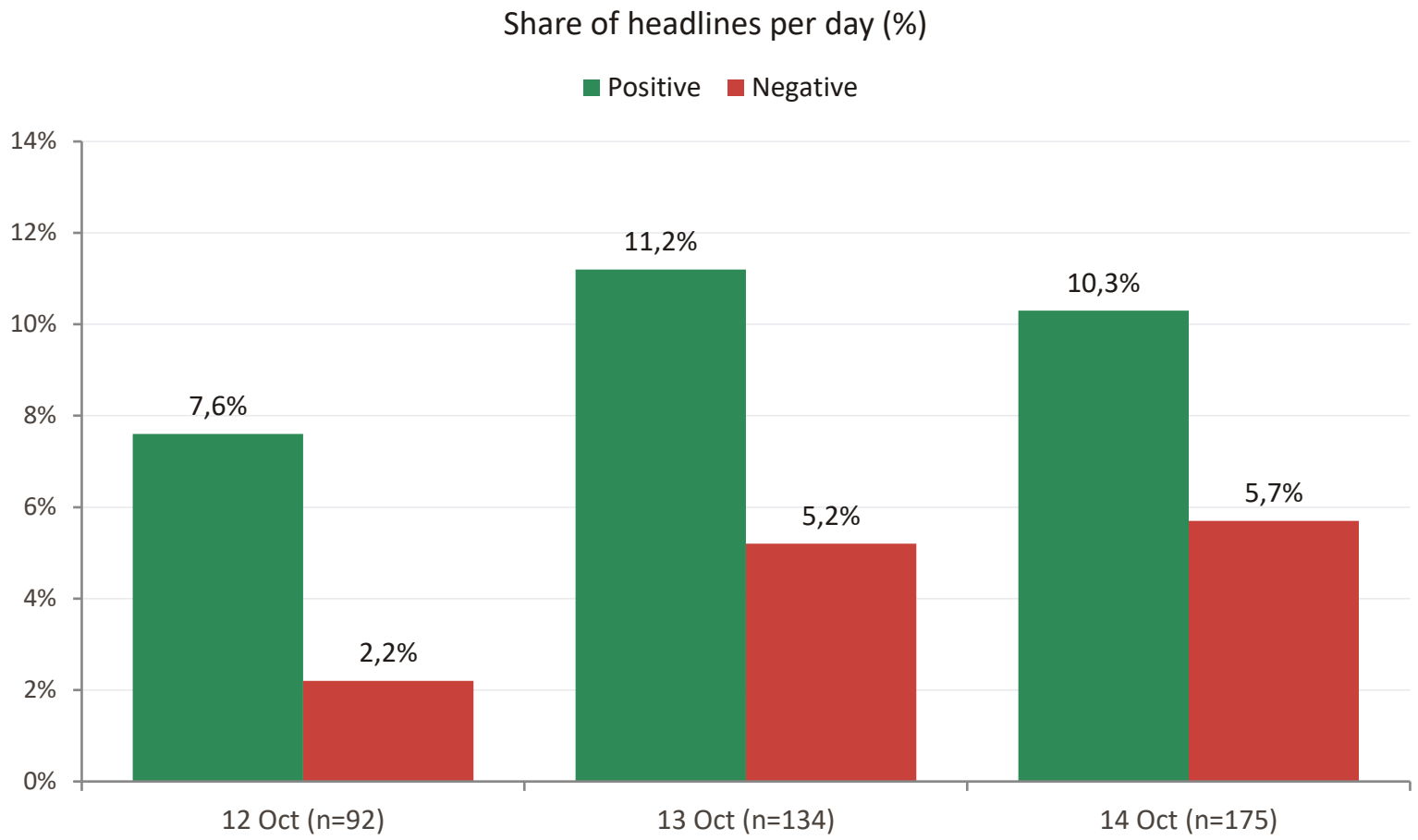
Model: cardiffnlp/twitter-roberta-base-sentiment-latest (Hugging Face, run with transformers on our own CPU). RoBERTa-base trained on ~124M tweets, English, 3 labels. **Why:** built for short texts like headlines and has a neutral class — most news is neutral.

401 headlines in 54 hours, negative share rising

85% **neutral**
342 headlines

10% **positive**
40 headlines

5% **negative**
19 headlines



Own data: solarmood.csv, publish times 12 Oct 10:31 – 14 Oct 18:30 (CEST). 12 and 14 Oct are partial days; the n per day is shown next to every bar.

Behind the dip: one story, three outlets

14 Oct, 08:46 – 10:28 · 4 negative headlines in under 2 hours

08:46	China solar giants shed another 40,000 jobs last year as overcapacity persists	Reuters
09:07	China solar giants shed another 40,000 jobs last year as overcapacity persists	The Economic Times
09:28	China's Solar Giants Cut 40,000 More Jobs as Overcapacity Crisis Persists	China-Global South Project
10:28	PNE cuts 2026 outlook on project delays	Renewables Now

1

Real signal. Chinese overcapacity pushes panel prices and Dutch installers' margins: a theme worth a position.

2

But copies move the line. 3 of the 4 negatives are the same Reuters story. We count headlines, not stories.

Own data: solarmood.csv. Dashboard filter: sentiment = negative, hour = 14 Oct 06:00 and 07:00 UTC

Honest test: the model agreed on 21 of 50

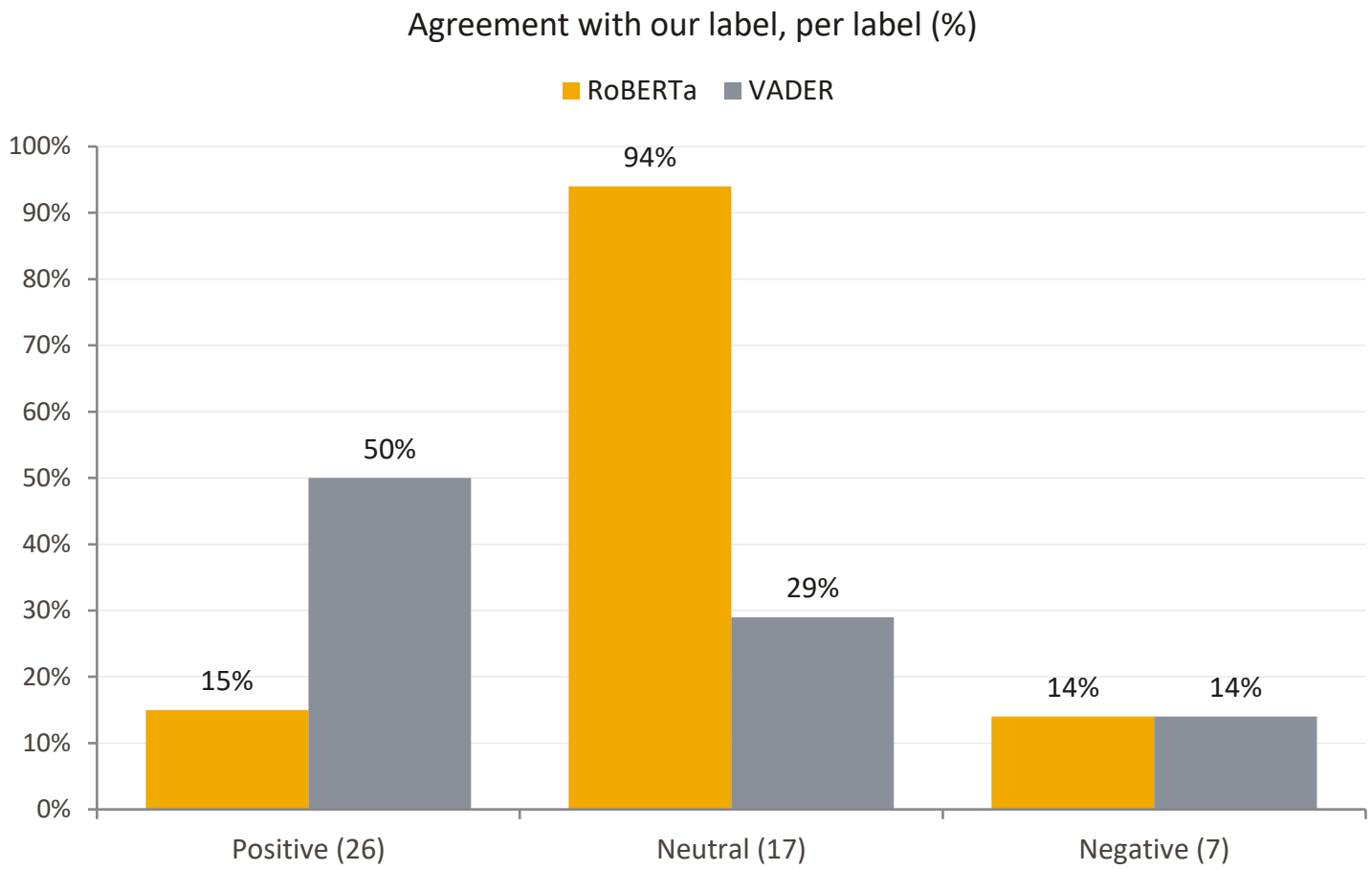
42%

RoBERTa (our model)
21 / 50 agree

38%

VADER (word-list baseline)
19 / 50 agree

Our label rule: "Is this news good, bad or neutral for the solar sector?" Only 9 headlines were right in both models.



Own test: 50 headlines from our stream labelled by us (model_comparison_results.csv). Numbers in brackets = how many headlines we gave that label.

Where it goes wrong: tone is not impact

1 Double negation

"Three in four Germans oppose cuts to rooftop solar subsidies"

Us: positive · RoBERTa: neutral · VADER: neutral

Why: "oppose" + "cuts" = good news for solar, but both words sound negative

2 Word, not meaning

"Arkansas is trying to solve one of solar's biggest problems using a local rice farm"

Us: positive · RoBERTa: neutral · VADER: negative

Why: the word "problems" wins over the solution

3 Sector jargon

"Trump clean energy tax credit cutoff drives project rush as prices set to soar"

Us: negative · RoBERTa: neutral · VADER: positive

Why: "rush" and "soar" sound upbeat; losing tax credits hurts the sector

4 Our own mistake

"Buyer beware: going solar turned into a 'nightmare' for a Richmond man"

Us: positive · RoBERTa: negative

Why: the model was right. The test also tested us: we need a second labeller

Also live: "Battery energy storage failures are down 98% since 2018" was labelled negative (0.80) because of the word "failures".

We pick RoBERTa: a filter, not a verdict

RoBERTa

- 16 of 17 neutral headlines right: few false alarms
- Its negatives are mostly real problems (defects, job cuts, fires)
- Misses good news: found 4 of 26 positives

VADER (word list)

- Finds more positives: 13 of 26
- But called 11 of 17 neutral headlines positive
- Counts words, so it cannot handle negation or jargon

→ **Why not just read the news yourself?** ~180 headlines a day takes about an hour (20 s each). SolarMood flags ~8 negative headlines a day to read in 3 minutes. Next step: fine-tune the model on our sector-impact labels, so it learns "good for solar" instead of "sounds happy".

Own calculation: 401 headlines and 19 negatives in 54 hours → ~178 and ~8 per day.

Risks specific to SolarMood, and what we did

1	**English, US-weighted feed**	The line is not Dutch opinion. Holland Solar could miss local resistance to a Dutch solar park.	Done: source and language stated in the README and here. Next: Dutch feed (hl=nl) and a multilingual model.
2	**Model reads tone, not impact**	It missed 22 of our 26 positive headlines, so the mood looks gloomier than the news is.	Done: warning box on the dashboard and the test results shown to the user.
3	**Copies and off-topic items**	One Reuters story counted 3×; The Cool Down (19 items) also posts about snakes and foggy windows.	Done: duplicate titles skipped. Next: group copies into one story and filter off-topic items.
4	**Small numbers, real names**	1–25 headlines per hour: one item swings the %. Headlines can name private people.	Done: article counts shown on the dashboard; we store only headline, time and label, no links, authors or profiles.

Not for: investment decisions, judging one company or person, or claiming what "the public" thinks about solar.

Live demo

solarmood-milou-jkelt2tjctww4cb8fjsbri.streamlit.app

1. Total articles and the sentiment bar chart
2. Sentiment over time: the hourly line chart
3. Filter: negative + 14 Oct 06:00 UTC → the China job-cut headlines
4. Model comparison and the warning box

What we learned

SolarMood turns ~180 headlines a day into ~8 worth reading — but it measures tone, not impact on the sector.

42%

agreement with our labels

Next: fine-tune on sector-impact labels · add Dutch news · group copied stories · second labeller